

Hybrid Search

Sparse Vectors

+

Dense Vectors

RRF
Rank Reciprocal Fusion

BM25 → Sparse Vectors

OPENAI → Dense Vectors

Combining

Weighted sum of normalized scores

α * vector score + $1 - \alpha$ * bm25 scores

Normalization

S Min Max Normalization BM25 scores
D Invert & normalize vector scores

Normalize scores

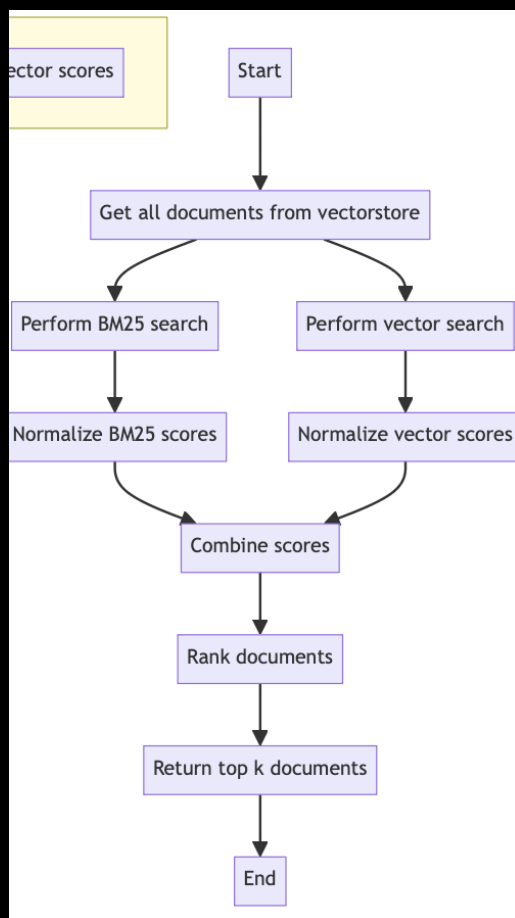
Min-max normalization for BM25

Invert and normalize vector scores

Combine scores

Weighted sum of normalized scores

α * vector_scores + $1 - \alpha$ * bm25_scores



Focus:- Better Quality Sparse Vectors

TF-IDF, BM25

- 1) Document length
- 2) Keyword Saturation

Static → Formula
↘
learnable

SPLADE

BERT

MLM

Learnable Sparse Vector Masked Language Model

Tokenatization

Token

Fill in the blanks

Query $\rightarrow$ Sub word level Tokenization

Predict the most preferable token.

MLM

BERT Vocab Size:- 30,552 tokens

Context of the entire Sequence.

Splade will combine

1) Probabilities :- across each token

Encoder

prefer

$\downarrow$

probable

Logarithmic Saturation

Logic 0
Constant

↓
Data sparsity

Dim:- 30,572

1000, 30000

Learnable Sparse Vectors
↑
Static

RRF

Reciprocal Rank Fusion

Dense:- Semantic

Sparse:- BM25

Merging both the results?

Both will return list of ranked items.

Two Ways

1) Rank

2) Scores

RRF $\rightarrow$ play ranks not with score

Formula,
$$RRF = 1 / (K + rank)$$

$$K = \text{constant} = 60$$

A Document

$$\text{list [A]} = 1 / (60 + 1) = 0.0164$$

$$\text{list [B]} = 1 / (60 + 5) = 0.0154$$

$$\begin{aligned} \text{Combine} &= 0.0164 + 0.0154 \\ &= 0.0318 \end{aligned}$$

Benefits :- *

- * No Tuning Required *
- * Score agnostic
- * No Domination of a single method

Rank	Dense	Sparse
		Doc C
1	Doc A	Doc A
2	Doc B	Doc D
3	Doc C	Doc B
4	Doc D	

Doc A RAF

$$D = 1 / (60 + 1) = 0.0164$$

$$S = 1 / (60 + 2) = 0.0161$$

$$\gamma_{\text{total}} = 0.0325$$

Doc B

$$B, \text{RAF} = 0.0317$$

Doc C

$$C, \text{RAF} = 0.0323$$

Doc D

$$RRF, D = 0.03125$$

Final Fused Rank

Doc A	=	0.0325
C	=	0.0323
B	=	0.0317
D	=	0.03125

RSF $\rightarrow$ Relative Score Fusion

play with score

$\times$ Rank.

Raw Scores:-

$$D_{vec} = [$$

$$S_{vec} = [$$

]

]

Not directly comparable

Normalization (0 - 1)

$$n_score = (score - min_score) / (max_score - min_score)$$

Sum the normalized across all utterances

Row Results

Doc	Dense	Sparse
		Doc C (18.5) C
Doc A (0.92)		Doc A (14.2) A
Doc B (0.85)		Doc D (10.8) D
Doc C (0.71)		Doc B (7.1) B
Doc D (0.68)		

Dense: Min = 0.68 Max = 0.92

$$\text{Doc A} = \frac{(0.92 - 0.68)}{(0.92 - 0.68)} = 1$$

$$\text{Doc B} = \frac{(0.85 - 0.68)}{(0.92 - 0.68)} = 0.71$$

$$\text{Doc C} = 0.125$$

$$\text{Doc D} = 0$$

$$\text{max} = 18.5$$

$$\frac{\text{Sparse}}{\text{min} = 7.1}$$

$$\text{Doc C} = \frac{(18.5 - 7.1)}{(18.5 - 7.1)} = 1$$

$$\text{Doc A} = 0.62$$

$$\text{Doc D} = 0.32$$

$$\text{Doc B} = 0$$

<u>New Table</u>			<u>Sparse</u>
	RSF	<u>Dense</u>	
Doc A	1.62	1	0.62
C	1.125	0.125	1
B	0.71	0.71	0
D	0.32	0	0.32

Sum of normalized scores

Compare RRF & RSF

Situation:-

when scores are reliable
RSF

when scores are not reliable
RRF

Outliers in data

RRF X

RSF ✓

10 chunks

c_1, c_2, c_3, b_u, b_s

Rank	Dense	Sparse
	c_3	c_7
1		c_3
2	c_5	c_9
	c_1	

ReRanker

Hybrid Search

Embedding Model

Bi-encoders

Query

Context / Docs

Problems :-

1) less Precision

* Recall *

2) Missing Context

Why ?

Recurrent are cross encoders.

Joint Encoding of query & doc.

capture deeper relationships. and context.

* Self-Attention *

Recurrents :- Query intent with the relationship of document context.

Reranker :- $\rightarrow$ Cross Encoder
LLM Reranker.

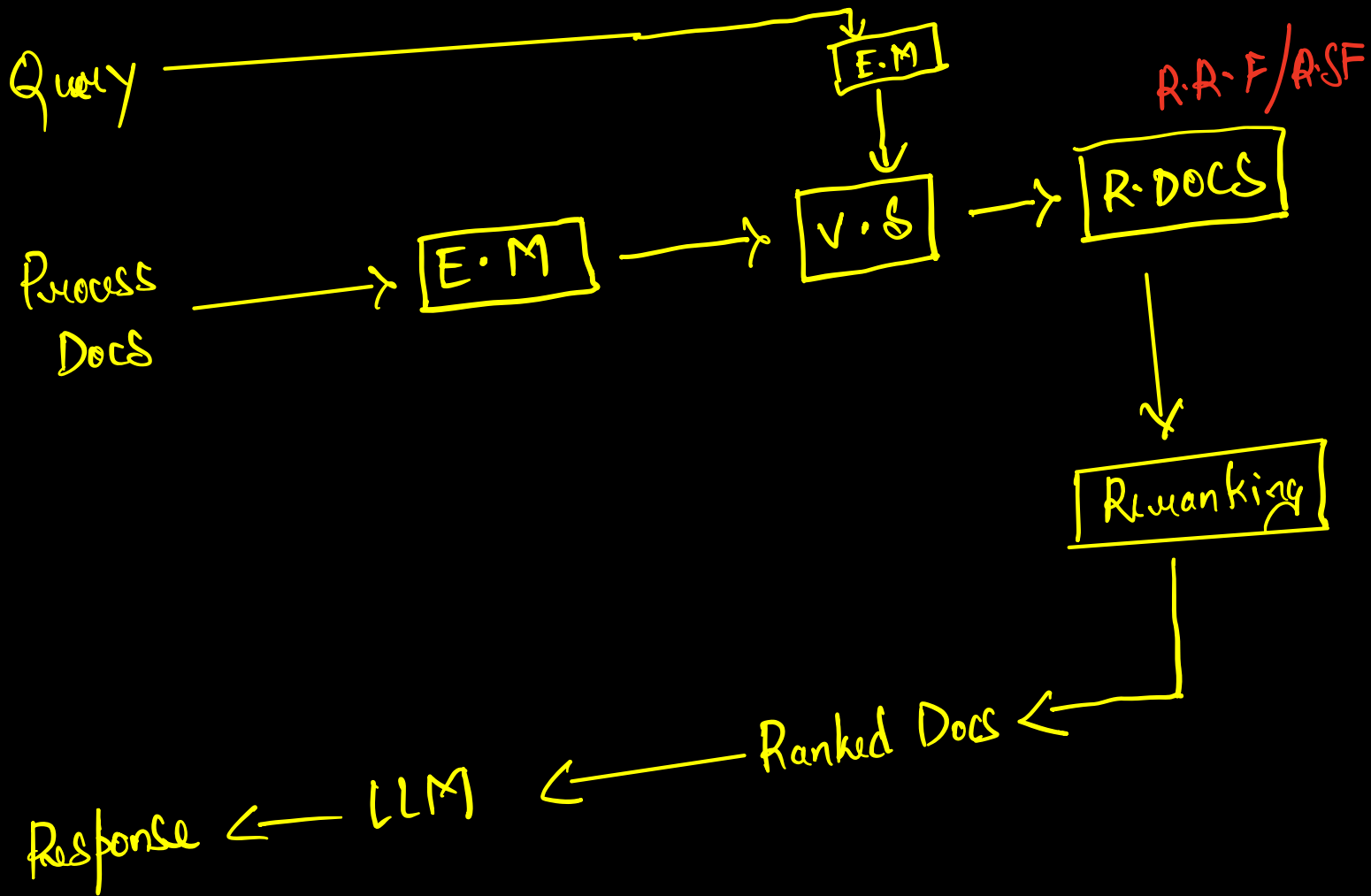
$$C.E(R_L) = (a, d)$$

During inference time.

Reranker:- 2 stage retrieval system.

Vector Search $\rightarrow$ Reranking
Results

{ length of query }



Top = 50 ~ 1000
Reranking = { 20 - 25 }

Bigque Rerankers

High accuracy (deep semantic relationship)

latency high

Cohere / Voyage AI

Short Query $\rightarrow$ Rerank2-lite

Long Query $\rightarrow$ Rerank-2

Chunk Size :- not very small

Use :- (800 ~ 2000)